

INCIDENCE RATE AND DISTRIBUTION OF PAEDIATRIC NOSOCOMIAL INFECTIONS

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ABSTRACT

The mechanism of action of Nosocomial infection as in any other infectious disease is dependent on host, agent and environment factors. Risk factors for the host are age, nutritional status and co-existing disorders. The organism's intrinsic virulence as well as its ability to colonize and survive within institutions influence nosocomial infections. Diagnostic procedures, medical devices, medical and surgical treatments are risk factors in the hospital environment. Unlike the adults, children are the more susceptible subjects of such kind of infections especially the neonates. Neonatal intensive therapy units usually have a prevalence of about 20% and superimposed by virulent and usually antibiotic resistant strains because of following reasons:

- Children that are admitted usually have weak immune system, so they may acquire the colonization of any type of organism.
- Ill babies require more handling hence the spread via contact.

In describing the incidence rate and distribution of nosocomial infections in children, for implementation and improving infection control measures, one hundred and twenty four isolates of *Pseudomonas aeruginosa*/*Pseudomonas* species, *Staphylococcus aureus* (MRSA/MSSA) and *Klebsiella* specie, were collected from a pediatric hospital setting in Karachi.

The results showed more than 24 % (124 isolates of 513 cultures that were suspected of acquiring nosocomial infection) incidence percentage rate. Nosocomial infections are caused in children in a developing country due to the lack of sanitation, monitoring and implementation of infection control procedures. Common sources of such infections were the water supply, poor ward/operation theatre hygienic conditions and utensils.

INTRODUCTION

Nosocomial infections are disseminated by airborne spread; Tuberculosis, Water-borne spread, Food acquired spread, contact spread (Albers *et al.*, 2001); microorganisms on the hands of staff may be resident (present but not removed by washing) or transient (recently acquired from another source), Inoculation mode of spread; blood transfusion and tissue donation, infection by accidental inoculation, infection from contaminated infusion fluids are the examples of infections spread by inoculation in a Hospital. Spread by Indwelling medical devices; cannulae used in different admixtures and drug administrations

led to difficulties with bacterial colonization and resultant infection (Maki *et al.*, 1973; Patel, 2000; Tullu, 1998). Nosocomial Infections cause considerable morbidity and mortality and are associated with prolonged hospital stay and increased health care costs (Urrea *et al.*, 2004). The risk of infections in these patients remains high because invasive devices allow organisms to bypass normal host defenses. Additionally, this patient group often has coexisting metabolic or organ system dysfunctions. Antibiotic pressure has led to the development of drug-resistant organisms within the PICU, thereby causing infections that are increasingly difficult to control (Rowin *et al.*, 2003).

MATERIAL AND METHOD

The isolates were maintained in nutrient agar slants and were kept in refrigerator as soon as they were transported.

Identification requirements:

The biochemical media Urease agar, Citrate agar, and differential culture media. This includes the microscopy, sugar broths like Mannitol and for *Pseudomonas* and cultural characteristics.

Media used for identification:

MacConkey agar (Oxoid), Tryptone Soy agar TSA (Oxoid), Triple sugar iron agar TSI (Merck), Blood agar base (Oxoid):

- Isolates along with the control specimens were brought to room temperature and then by the help of wire loop or straight wire, are inoculated in Mueller Hinton broth tubes.
- The tubes are then incubated at 37°C for at least 6 hours.
- Petri dishes for sensitivity are labeled with the isolate code respectively and by the help of a cotton swab are spread (lawn preparation) on the agar.
- Prescribed antibiotic discs are then dispensed and placed on to the bacterial lawn by the help of a sterile forcep.
- Petri plates are incubated at 37°C for 24-36 hours.
- Interpretation of the zone of inhibition is then made carefully by the ruler. Appropriate interpretive zone diameters were recorded as per National Council of Clinical Laboratory Standards USA (NCCLS).

EXPERIMENTAL

Collection of specimens:

One hundred and twenty four isolates of *Pseudomonas aeruginosa*, *Pseudomonas sp.* *Staphylococcus aureus* and *Klebsiella sp.* were collected from children of age from as minimum as 8 months to as maximum as 12 years of different pediatric hospital settings of

Karachi during the six months (as per from 20-07-01 to 29-01-02).

- Morphological, cultural and biochemical methods further identified these isolates.

Data collection for Incidence:

- The number of admissions was collected during the period of study with the infection status of the patient at day zero for the determination of incidence rate. This includes the blood complete picture, infective status of the patient by virtue of its febrile status, and pus formation, duration of its stay and the duration after which patient specimen was evaluated for infection.

Environmental Check:

Environmental studies were also conducted and evaluation of different utensils used in such setups was made as follows:

- At floor level, Bed level and AC level (By leaving open culture plates for a while).
- The utensils used by the staff, OT apparatus, feeding and cleansing utensils etc. (By the help of cotton swabs etc).

The staff members were also examined by taking their hand impressions before and after workout.

- In order to determine resistance pattern disc diffusion method (Kirby Baur) (Oxoid susceptibility disc of 30µGM) was used. The usual sources of contamination were mentioned.

RESULTS AND DISCUSSION

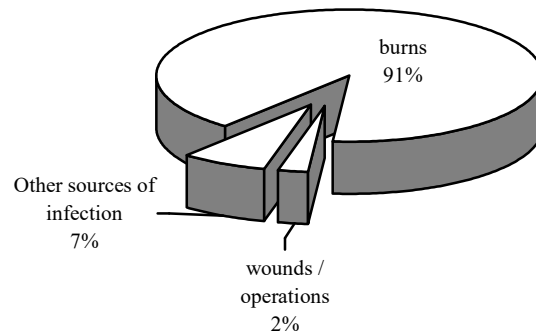
The children that have been included in this study (n=124) were those patients who at least have been hospitalized for seven or more days. Most of them were burnt children (91%, n=112)) and those who were kept in incubators or exposed to hospital/surgical utensils etc. and/or came with any complaint and there culture and sensitivity specimen sent after at least 24 hours of there hospitalization (7%) and those who have been operated at different

settings (2%) (Figure 1, Table 1). It has been observed that most of colonization and infections occurred within a week of admission i.e. 76 % (n=94) (Figure 2, Table 2). This may be due to the fact that burn children have weak immunity (Gastmeier, 2002)

incidence of nosocomial infection was 1.5 per 100 patient-days (Urrea *et al.*, 2003). The rate of PICU-acquired colonization, in a study, was 12.5% (25), the rate of PICU-acquired infected patients was only 2% (4 events), and the infections rate was 3.5% (7 events) (Pigna *et*

Table 1
Distribution of % Nosocomial Infection Conditions

Burns	90.32 (n = 12)
Wounds/operations	02.41 (n = 3)
Other sources	07.25 (n =9)



□ burns □ wounds / operations □ Other sources of infections

Fig. 1: Figure showing distribution of Pediatric Nosocomial Infections.

against the organisms associated in nosocomial infections and the conditions that should be available for them to prevent such infections were lacking, so they were most susceptible to infections in the first week of stay. In a group of 41 children with a mean total burn surface area (TBSA) of 18.9% 42 hospital-acquired infections were observed. The overall infection rate was 59.7 nosocomial infections per 1000 patient days, the device-associated nosocomial infection rates per 1000 (Gastmeier, 2002). A nosocomial infection rate of 3.15% was observed among the 412 neonates hospitalized for more than 2 days in the pediatric intensive care unit and a special care baby unit (Boutte *et al.*, 1990). Incidence of nosocomial infections was 16.86% in a surveillance study (Khilnani *et al.*, 2004). The

al., 1995). In a study, the incidence rate of pneumonia and bacteremia/sepsis was 28 and 26 cases respectively, per 1000 days of exposure to an invasive device (Martinez-Aguilar *et al.*, 2001). The infection rate in the present study shows more than 24%, 124 isolates of 513 cultures that were suspected of acquiring nosocomial infection for the mentioned period. Causative agents in nosocomial infections included *Staphylococcus aureus*, coagulase-negative *Staphylococcus* and *Escherichia coli* (Allpress *et al.*, 2004). In another study, the main isolated agents were coagulase-negative *Staphylococci*, *Pseudomonas aeruginosa*, *S. aureus*, *K. pneumoniae*, and *Candida* spp. (Chang *et al.*, 2003). Coagulase-negative staphylococci and *Pseudomonas aeruginosa* were the most

Table 2
Distribution of cases According to Hospital Stay

Duration at which culture came positive (Days)	No. of cases	% age (No. of cases/124 *100)
1 – 7	94	75.80
8 – 14	27	21.77
15 – 21	03	2.41

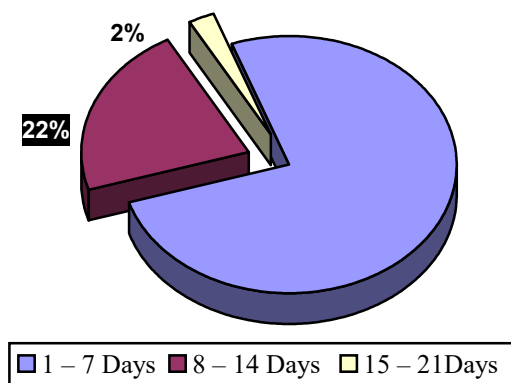


Fig. 2: Figure shows the duration of stay at which culture came positive

common organisms isolated in a study (Urrea *et al.*, 2003). Coagulase-negative *Staphylococci* were the most common isolates in bacteremias in a study (Urrea *et al.*, 2004). In the present study children of age group from 8 months to 12 years were included. In all, 124 isolates of suspected nosocomially infected children (infected by *Pseudomonas aeruginosa*, *Pseudomonas sp.*, *E. coli*, *Klebsiella pneumoniae*, *Kl. sp.*, *Proteus mirabilis*, *Citrobacter sp.* and *Morganella sp.*) were included (Table 3, Figure 3). The risk conditions that were most closely related to NI acquisition in a study, were: prolonged hospital stays (69.4%), prematurity (60.9%) and exposure to high-risk device procedures (95.4%) (Chang *et al.*, 2003). Nosocomial Pneumonia (NP) developed in patients undergoing Mechanical Ventilation (MV) in a surveillance study and the main risk factors for developing NP were duration of MV and duration of stay in the PICU (Tullu *et al.*, 1998). The most frequent pathogens associated with ventilator associated pneumonia according to a study are *S. aureus* (24.1%),

followed by *P. aeruginosa* (16.8%) and *Klebsiella* spp. (12.1%) (Gastmeier *et al.*, 2003). The overall mortality attributable to pediatric nosocomial infections has been estimated at 11%. This report describes the epidemiology of nosocomial infections in children within PICUs in the United States that use the standard surveillance protocols and nosocomial infection site definitions from the National Nosocomial Infections Surveillance (NNIS) System of the Centers for Disease Control and Prevention (Michael J. Richards *et al.*, 1999). The best current approach for preventing PICU infections centers on consistent hand washing between patients, early discontinuation of invasive devices, and appropriate isolation strategies (Rowin *et al.*, 2003). Among the miscellaneous causative agents for PNIs that have been recovered in this study, were the spp. of genus *Citrobacter* (1%), *Morganella* (1%) and *Proteus* (4%). The major sources of such organisms are water, air and utensils of the ward, O.Ts and utensils (Table 4).

Table 3
Distribution of Organisms in PNIs ($x = 124$)

Organism	Isolates	%age
<i>Pseudomonas aeruginosa</i>	62	50.00
<i>Pseudomonas sp.</i>	06	4.83
<i>Klebsiella pneumoniae</i>	16	12.90
<i>Klebsiella sp.</i>	02	01.61
<i>Staphylococcus aureus</i>	28	22.58
<i>E. coli</i>	03	02.41
<i>Proteus mirabilis</i>	05	04.03
<i>Citrobacter sp.</i>	01	0.806
<i>Morganella sp.</i>	01	0.806

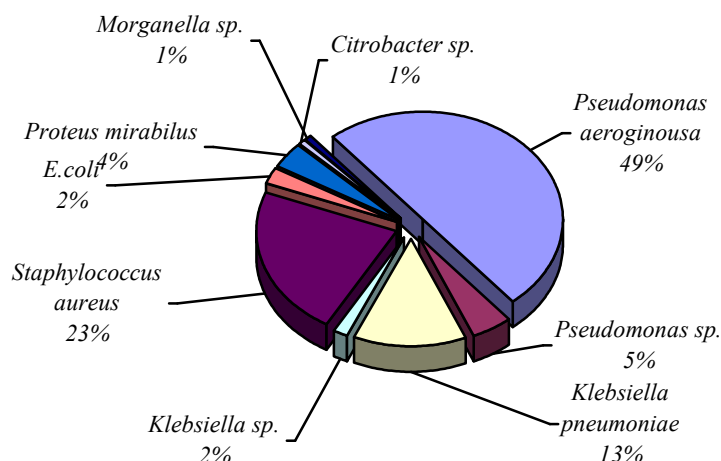


Fig. 3: Distribution of Organisms in Pediatric Nosocomial Infections.

Of the 124 isolates, 91% (112) were burn cases i.e., children that have been admitted with some kind of burn conditions. This is also true for the children that have been admitted in general wards where there were very least hygienic and isolation conditions, 7% (9) were of other infections and 2% (3) from O.Ts (Table and Figure 1).

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Table 4
Table showing organisms recovered from different hospital sources

Environmental check for PNIs	
Sources	Organisms
Water Supply Tap water Boiled water	<i>Pseudomonas sp.</i> , <i>Bacillus sp.</i> , <i>Klebsiella pneumoniae</i> <i>Acinetobacter sp.</i>
Wards Surgical Floor level Bed level A/C level	<i>Klebsiella sp.</i> , <i>Staphylococcus aureus</i> , <i>Bacillus sp.</i> <i>Klebsiella sp.</i> , <i>Staphylococcus aureus</i> , <i>Bacillus sp.</i> ----
OT Floor level Bed level A/C level	<i>Staphylococcus aureus</i> (MSSA), <i>Bacillus sp.</i> <i>Staphylococcus aureus</i> , <i>Bacillus sp.</i> <i>Staphylococcus aureus</i> , <i>Bacillus sp.</i>
Burns Floor level Bed level A/C level	<i>Staphylococcus aureus</i> (MSSA), <i>Bacillus sp.</i> <i>Pseudomonas sp.</i> ----
Utensils Ward utensils Pyodine container Normal saline cup Ventilator bowl	No growth <i>Staphylococcus aureus</i> , <i>Bacillus sp.</i> <i>Pseudomonas aeruginosa</i>
OT utensils Anesthesia trolley Laryngoscope	<i>Staphylococcus epidermidis</i> , <i>Bacillus sp.</i> <i>Staphylococcus epidermidis</i>
Equipments Incubators Ventilators Ventilator tip Ventilator tube Suction tube Ambo. bag	<i>Staphylococcus aureus</i> (MSSA), <i>Bacillus sp.</i> <i>Pseudomonas aeruginosa</i> , <i>Klebsiella sp.</i> , <i>Kleb. pneumoniae</i> , <i>E.coli</i> <i>Staphylococcus aureus</i> (MRSA) <i>Staphylococcus aureus</i> (MRSA) <i>Pseudomonas aeruginosa</i> , <i>Klebsiella sp.</i> , <i>Staph. aureus</i> (MRSA) <i>Staphylococcus aureus</i> (MRSA)
Staff Neonatal ICU Burns	<i>Staphylococcus aureus</i> (MRSA), <i>Bacillus sp.</i> <i>Klebsiella sp.</i> , <i>Pseudomonas sp.</i> , <i>Ps.aeruginosa</i> , <i>Bacillus</i> , <i>MSSA</i>

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